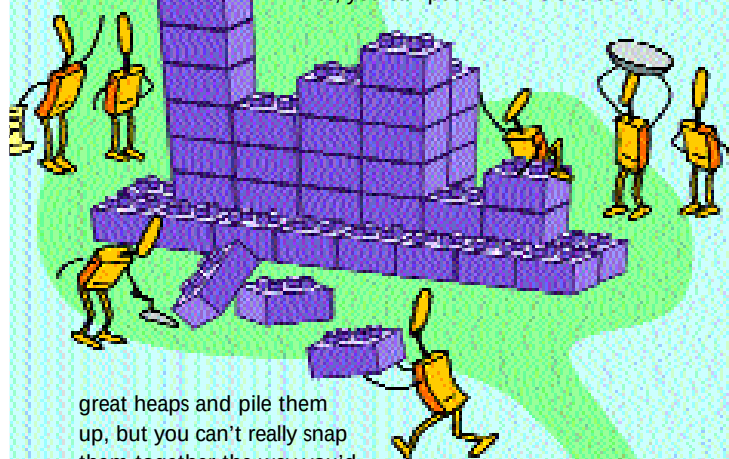


NANOBOTS

Nobel prize-winning physicist Richard Feynman said in 1959, "The principles of physics, as far as I can see, do not speak against the possibility of manoeuvring things atom by atom." This is essentially the principle behind molecular robotics, or molecular nanotechnology, which might ultimately produce nanobots. Dr Ralph Merkle, one of the pioneers of nanotechnology, says that the engineering we do today—grinding objects, or building bicycles, or even lithography, is like trying to build something out of LEGO blocks with boxing gloves on. "Yes, you can push the LEGO blocks into



great heaps and pile them up, but you can't really snap them together the way you'd like." With nanotechnology, he says, we will be able to fabricate "an entire new generation of products that are cleaner, stronger, lighter and more precise." Molecular nano aims at no less than controlling the assimilation of things—computing machines, robots, or whatever—atom by atom. Here is where nanobots come in—if someone or something has to fabricate a device at the atomic level, it would have to be something really small, something measurable in nanometres. Also, it would be essential for fabricated devices to self-replicate to keep costs down: The nanobot will manufacture one device, which will self-replicate as many times as one wanted it to, thereby creating a collection of those devices. Here's where the gray goo comes in.

Merkle, along with Eric Drexler—the author and researcher who pioneered the field of nanotech—envisioned a long time ago that a mechanical nanocomputer would operate somewhat like a vastly scaled down and programmable version of a mechanical calculator. But why do we need nanomanufacturing? What would its impact on the computing industry be? Smaller, faster computers, of course. Merkle says that molecular manufacturing "should let us make computers at a manufacturing cost that's less than a dollar per pound, operating at frequencies of tens of gigahertz or more, with linear dimensions for a single device of roughly 10 nanometres, high reliability, ..."

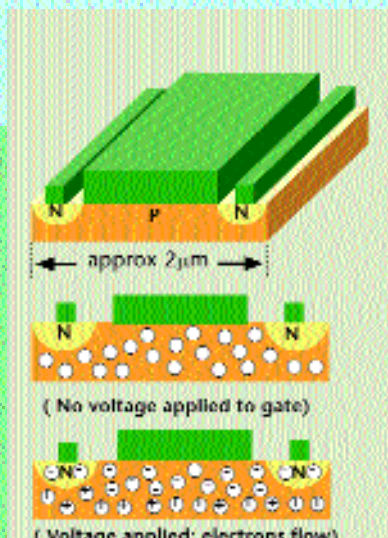
The mechanical approach to nanocomputing has been severely criticized by many. It also happens to be reminiscent of Charles Babbage's famous Analytical Engine, which many call the first computer.

TODAY'S TRANSISTORS

Today's computers represent, and manipulate, 1s and 0s using physical devices measured in microns. (So we're still in the 'microworld', as opposed to the 'nanoworld') These devices are transistors, and the trick involved in shrinking computers has been, mostly, to make transistors smaller and smaller, and pack them more closely together.

What is fundamental and useful about transistors is the fact that they can be put in one of two states—a '0' or a '1'. Most transistors used in modern computers are MOSFETs—a MOSFET is basically like a gate that's either open or closed. See figure below: when no voltage is applied to the gate (in green), electrons do not flow between the two 'N' regions, there's no current, and the transistor is 'off'—representing a 0.

When a voltage is applied to the gate, the 'electron carriers' in the 'P' region allow a current to flow between the two 'N' regions, and the MOSFET is 'on'—representing a 1.



The MOSFET keeps getting smaller over the years, but there's a limit on how small it can get for this mechanism to work. Three things—thermal voltage, heat and tunnelling—impose these limitations. Thermal voltage refers to the voltage that's around us all the time—25mV at room temperature, to be precise. As the transistor gets smaller, this thermal voltage effectively acts at a shorter distance, resulting in an increased electric field. When the electric field becomes high enough, the transistor can suffer an avalanche breakdown—a huge surge of electrons that cannot be controlled. And once that happens, the transistor goes kaput.

The second limitation on the transistor size is heat dissipation—that's what you use heat sinks for. The smaller the surface on which you apply, say, five volts, the more the heat generated. If it's not dissipated, it melts the chip. And there's no way to dissipate the kind of heat that will be generated by today's transistors if they were packed together on the nanoscale. The third problem is tunnelling, the quantum-mechanical process by which electrons flow between two adjacent transistors. If this happens, the transistors will no longer hold the bits they're supposed to.

If the problem is how to go about miniaturizing circuits further, down to the nanoscale, new methods must be used. Two of these are the resonance tunnelling diode (RTD), and the single-electron transistor (SET). Both operate using the tunnelling effect, and so, cannot suffer from the limitations imposed by the tunnelling effect. The RTD and the SET can therefore be built on a smaller scale than a regular MOSFET.